

Aligned paired circuits for a conditional $17 \cdot 2^{-63}$ saving

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This refinement uses Douglas Colkitt’s paired bit circuit and the retained interfaces of OpenAI’s *Integer multiplication below $n \log n$* , pinned at

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

The full multiplication theorem remains assumed.

Fix $h = 50$ and global pairs $\{0, 1\}, \dots, \{48, 49\}$. For common point i , order all points except i and $i \oplus 1$ increasingly, then append that partner. Run the unchanged local paired weighted-exclusion circuit in these coordinates. Relabelling preserves every local exclusion identity. All complete root pairs now agree across groups. Intern equal fixed-pair star sums using the existing compiler. The local graph still has 9813 additions and 1176 outputs; the merged graph has 435450 additions, 58800 partial outputs and $R = 494250$ roles.

Every retained sum still has a common point. The paired construction’s positive-definite source-span argument, reversible embedding, reverse complement frames and full auxiliary stage sharing therefore apply without additional rank loss. With $v = 19600$, $N = v^3$, $m = 125000$ and $L = 3v^2h^2$,

$$W = 2N + 2v^2(R + h) = 394839648000000, \quad s = Wm - N + 2L = 49354954232864000000.$$

Thus $D = Wm - s = 1767136000000$ and $\eta_b = 23/642375000$. Exact logarithm bounds give $\log m < 11737/1000$ and $\eta_b > (305/10^{11})(11737/1000)$, certifying $a = 305/10^{11}$. The complex motif and its saving $a_c = 1/10^{11}$ are unchanged.

Keep $\beta = 999/1000$, $\epsilon = 199/1000$, $\delta = 1/10000$, $C_1 = 2$ and set

$$\tau = 1 - a, \quad \sigma = 1 - a_c, \quad c = \beta a, \quad \lambda = 1 - (1 + \beta)a^2/2, \quad \lambda' = 1 - \beta a^2.$$

The paired stopped guard, Gaussian width and $b \geq 2^{40}$ cutoff are retained. All strict layer and setup inequalities pass exact rational comparison. The seven-margin minimum is

$$G = g_2 = g_3 = \epsilon\beta a^2 = \frac{739738521}{400000000000000000000000000} > 17 \cdot 2^{-63}.$$

This fixed positive gap absorbs the retained logarithmic factors and yields

$$T(n) = O(n(\log n)^{1-17.2^{-63}})$$

conditionally in the same fixed-tape model. The target improves the preceding 2^{-59} headline by 17/16. There is no measured practical speedup or unrestricted optimality claim.

The full dependency ledger is `docs/research/aligned-paired-network.md`. `certificates/aligned-paired-network.json` checks all full-size coefficients, frames, counts and parameter inequalities. Small independent support and dirty scratch tests complement the general transfer proof. The independent patch `patches/h50-aligned-paired.patch` updates all affected upstream statements. These checks do not formally verify the complete upstream theorem.